Section 3.6: Conditional Probability

Finite Mathematics MTH 107 Fall 2014

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/home/jim/JamesAshe/MHU/images/MarsHil

Example 3.6.1.

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$$= \frac{13 \cdot 12}{n(1^{\text{st}}\ \heartsuit)}$$

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$$= \frac{13 \cdot 12}{n(1^{\text{st}}\ \heartsuit)}$$

$$= \frac{13 \cdot 12}{13 \cdot 51} = \frac{4}{17}$$

Product Rule

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Conditional Probability.

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Probability of A if B has occurred.

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p(A \mid B) \cdot p(B) &= p(A \cap B) \\
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\end{aligned}$$

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$p(A \cap B) = p(A \mid B) \cdot p(B)$

$$p(\text{heart and heart are dealt}) =$$

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Product Rule

$p(A \cap B) = p(A \mid B) \cdot p(B)$

Example 3.6.4.

IF two cards are dealt from a full deck, find the probability that both are hearts.

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&= {}^{4}\!/_{51} \\
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\end{aligned}
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Using a Tree Diagram:

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Product Rule

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Example 3.6.5.

If two cards are dealt from a full deck, find the probability that both are hearts.

Using a Tree Diagram:

Problems

Problem 3.6.1. Find the probability using a tree diagram.

Of two cards dealt from a full deck exactly one is a spade.

Problems

Problem 3.6.2. Find the probability using a tree diagram.

Of three cards dealt from a full deck exactly one is a spade.

Problem 3.6.3. Use the provided data to complete the following

Age	Obama	McCain	Other/NA
18-29	11.9%	5.8%	0.4%
30-44	15.1%	13.3%	0.6%
45-64	18.5%	18.1%	0.4%
65+	7.2%	8.5%	0.3%

Problems

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